

Dizajn kompajlera - zadaća 3

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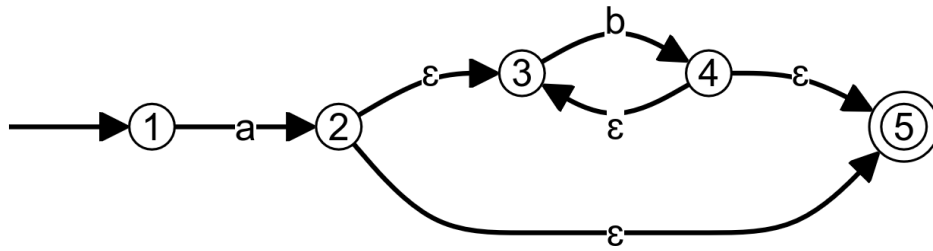
Zadatak 1

Regularni izraz:

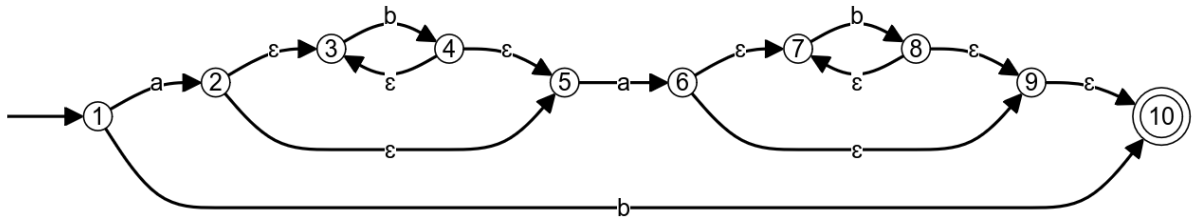
$b \mid (ab^*ab^*)$

pretvoriti u NFA, zatim u DFA, te odraditi postupak minimizacije DFA.

NFA za ab^* :



NFA za $b \mid (ab^*ab^*)$:



$$\varepsilon(1) = \{1\} = A$$

$$\varepsilon(MV(A, a)) = \varepsilon(\{2\}) = \{2, 3, 5\} = B$$

$$\varepsilon(MV(A, b)) = \varepsilon(\{10\}) = \{10\} = \underline{C}$$

$$\varepsilon(MV(B, a)) = \varepsilon(\{6\}) = \{6, 7, 9, 10\} = \underline{D}$$

$$\varepsilon(MV(B, b)) = \varepsilon(\{4\}) = \{3, 4, 5\} = E$$

$$\varepsilon(MV(C, a)) = \emptyset$$

$$\varepsilon(MV(C, b)) = \emptyset$$

$$\varepsilon(MV(D, a)) = \emptyset$$

$$\varepsilon(MV(D, b)) = \varepsilon(\{8\}) = \{7, 8, 9, 10\} = \underline{F}$$

$$\varepsilon(MV(E, a)) = \varepsilon(\{6\}) = \{6, 7, 9, 10\} = \underline{D}$$

$$\varepsilon(MV(E, b)) = \emptyset$$

$$\varepsilon(MV(F, a)) = \emptyset$$

$$\varepsilon(MV(F, b)) = \varepsilon(\{8\}) = \{7, 8, 9, 10\} = \underline{F}$$

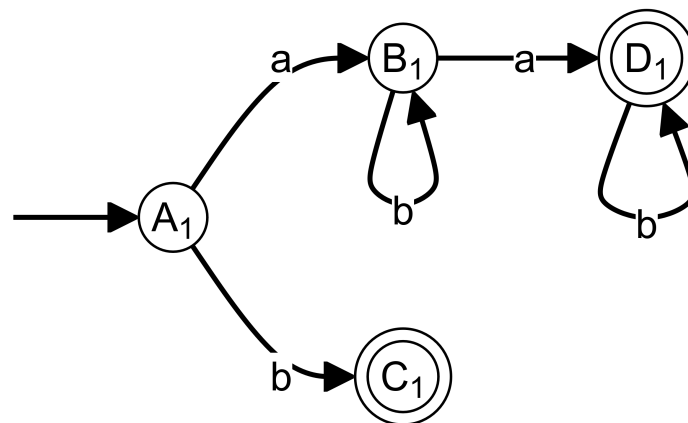
	a	b
$\rightarrow A$	B	<u>C</u>
B	<u>D</u>	E
<u>C</u>	$\emptyset$	$\emptyset$
<u>D</u>	$\emptyset$	<u>F</u>
E	<u>D</u>	$\emptyset$
<u>F</u>	$\emptyset$	<u>F</u>

$M_1 : (A, B, E)(\underline{C}, \underline{D}, \underline{F})$
 $M_2 : (A)(B, E)(\underline{C}, \underline{D}, \underline{F})$
 $M_3 : (A)(B, E)(\underline{C})(\underline{D}, \underline{F})$

$A_1 = A$
 $B_1 = B, E$
 $\underline{C}_1 = \underline{C}$
 $\underline{D}_1 = \underline{D}, \underline{F}$

	a	b
$\rightarrow A_1$	B_1	$\underline{C}_1$
B_1	$\underline{D}_1$	B_1
$\underline{C}_1$	$\emptyset$	$\emptyset$
$\underline{D}_1$	$\emptyset$	$\underline{D}_1$

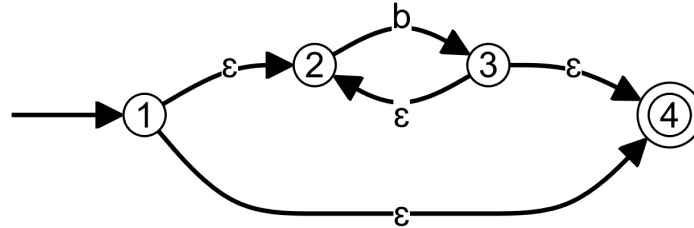
Minimalni DFA za $b \mid (ab^*ab^*)$:



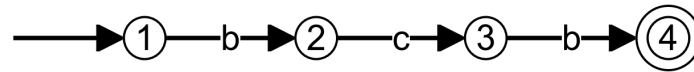
Zadatak 2

Regularni izraz: $a(b^*|bcb)$ pretvoriti u NFA, zatim u DFA, te odraditi postupak minimizacije DFA.

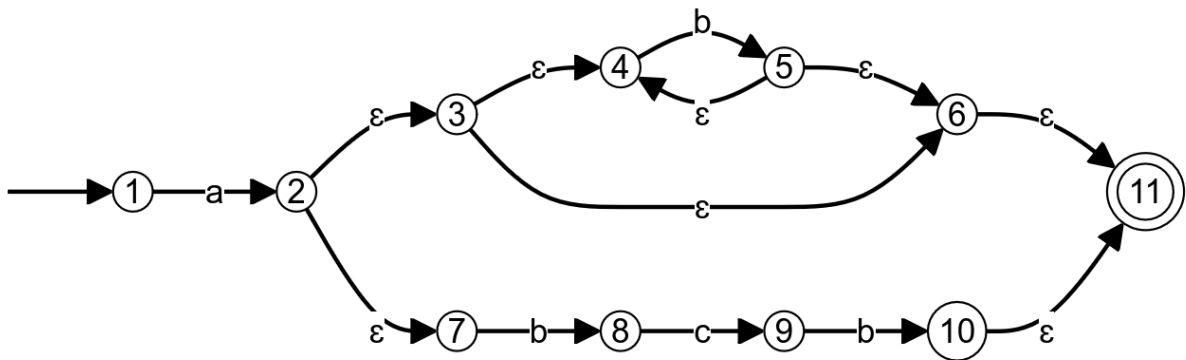
NFA za b^* :



NFA za bcb :



NFA za $a(b^*|bcb)$:



$$\varepsilon(1) = \{1\} = A$$

$$\varepsilon(MV(A, a)) = \varepsilon(\{2\}) = \{2, 3, 4, 6, 7, \underline{11}\} = \underline{B}$$

$$\varepsilon(MV(A, b)) = \emptyset$$

$$\varepsilon(MV(A, c)) = \emptyset$$

$$\varepsilon(MV(B, a)) = \emptyset$$

$$\varepsilon(MV(B, b)) = \varepsilon(\{5, 8\}) = \{4, 5, 6, 8, \underline{11}\} = \underline{C}$$

$$\varepsilon(MV(B, c)) = \emptyset$$

$$\varepsilon(MV(C, a)) = \emptyset$$

$$\varepsilon(MV(C, b)) = \varepsilon(\{5\}) = \{4, 5, 6, \underline{11}\} = \underline{D}$$

$$\varepsilon(MV(C, c)) = \varepsilon(\{9\}) = \{9\} = E$$

$$\varepsilon(MV(D, a)) = \emptyset$$

$$\varepsilon(MV(D, b)) = \varepsilon(\{5\}) = \{4, 5, 6, \underline{11}\} = \underline{D}$$

$$\varepsilon(MV(D, c)) = \emptyset$$

$$\varepsilon(MV(E, a)) = \emptyset$$

$$\varepsilon(MV(E, b)) = \varepsilon(\{10\}) = \{10, \underline{11}\} = \underline{F}$$

$$\varepsilon(MV(E, c)) = \emptyset$$

$$\varepsilon(MV(F, a)) = \emptyset$$

$$\varepsilon(MV(F, b)) = \emptyset$$

$$\varepsilon(MV(F, c)) = \emptyset$$

	a	b	c
$\rightarrow A$	<u>B</u>	$\emptyset$	$\emptyset$
<u>B</u>	$\emptyset$	<u>C</u>	$\emptyset$
<u>C</u>	$\emptyset$	<u>D</u>	E
<u>D</u>	$\emptyset$	<u>D</u>	$\emptyset$
E	$\emptyset$	<u>F</u>	$\emptyset$
<u>F</u>	$\emptyset$	$\emptyset$	$\emptyset$

Dobijena forma se ne može dalje minimizirati.

Minimalni DFA za $a(b^*|bcb)$:

